
Decision Trees, ROC/AUC analysis and Clustering Methods: A Data Mining Overview

CSE 435: Data Mining
Section: 221 D5

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Set 01: Decision Tree

What is a Decision Tree?

A Decision Tree is a visual model that helps us make decisions by mapping out different choices and their possible outcomes.

- **Tree-like Structure:** Starts with a main question (root node) and branches into possibilities
- **Supervised Learning:** Used for classification and regression tasks
- **Interpretable:** Easy to understand the decision-making process
- **Non-linear:** Captures complex relationships in data

Key Components

Root Node

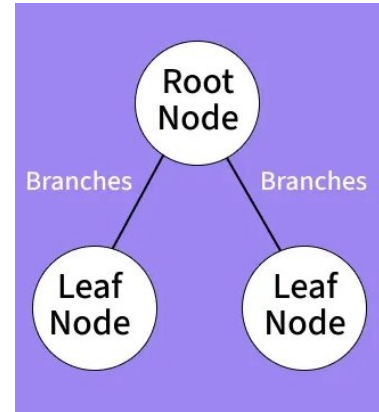
Starting point representing the whole dataset

Internal Nodes

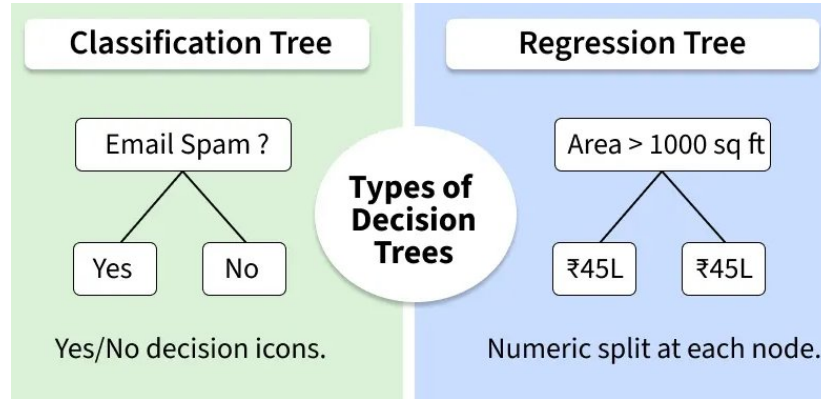
Points where decisions are made based on features

Leaf Nodes

End points where final decisions or predictions are made



Set 01: Decision Tree



Types of Decision Trees

Classification Trees

Purpose: Predict categorical outcomes

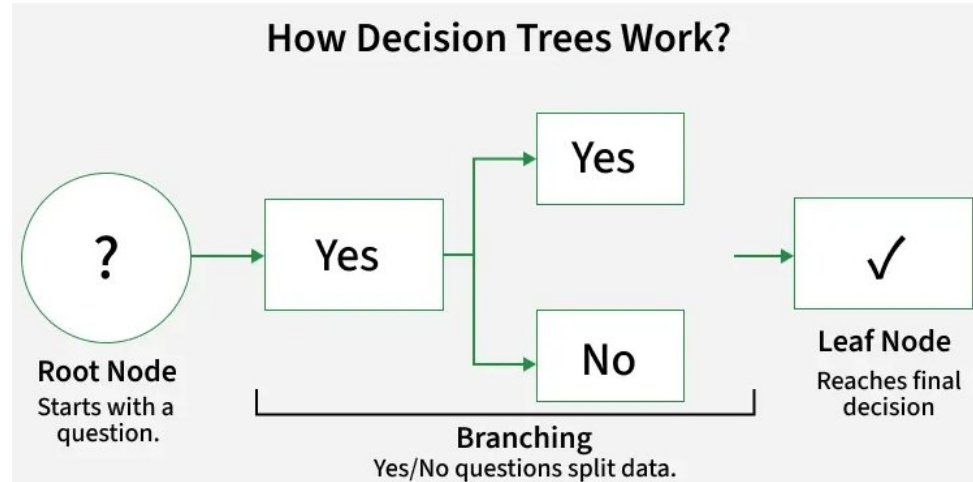
Example: Spam detection (spam/not spam), Disease diagnosis (diabetic/non-diabetic)

Regression Trees

Purpose: Predict continuous numerical outcomes

Example: House price prediction, Stock price forecasting

Set 01: Decision Tree



How Decision Trees Work?

- **Start with Root Node:** Begin with a main question about the dataset
- **Ask Yes/No Questions:** Tree asks series of binary questions to split data
- **Branch Based on Answers:** Different paths for yes and no responses
- **Continue Splitting:** Recursively divide data into subsets
- **Reach Leaf Nodes:** Final predictions when no more useful questions remain

Set 01: Decision Tree

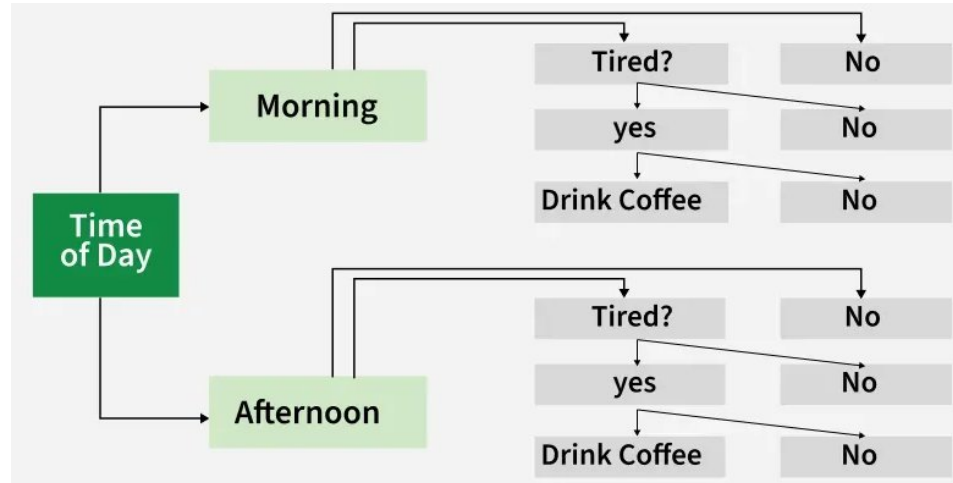
Real-World Example: Coffee Decision

Question: Should you drink coffee?

First Check: What time of day?

Morning: Ask "Tired?" → Yes = Coffee ✓ | No = No coffee ✗

Afternoon: Ask "Tired?" → Yes = Coffee ✓ | No = No coffee ✗



Set 02: concepts on Tree Splitting & Tree Pruning in Decision Tree

Split on Gender

Students = 30
Play Cricket = 15 (50%)

Female



Students = 10
Play Cricket = 2 (20%)

Male



Students = 20
Play Cricket = 13 (65%)

Split on Class

Class IX



Students = 14
Play Cricket = 6 (43%)

Class X



Students = 16
Play Cricket = 9 (56%)

What is Tree Splitting?

The process of partitioning a dataset into smaller and smaller subsets based on feature values.

Key Concept: At each node, the algorithm selects a feature and threshold that best separates the data into homogeneous subsets. This continues recursively until stopping criteria are met.

- Splits occur on a particular variable/feature
- Goal is to maximize homogeneity within each subset
- Process continues recursively down the tree

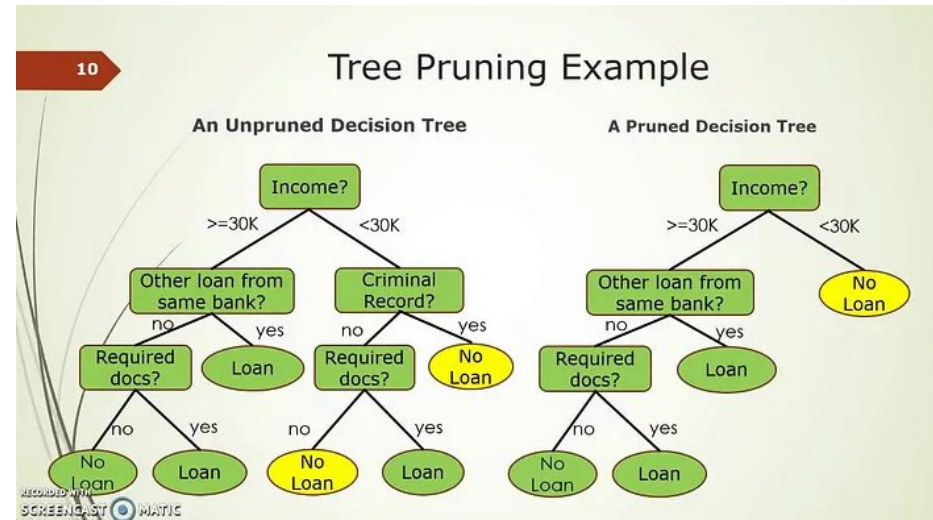
Set 02: concepts on Tree Splitting & Tree Pruning in Decision Tree

What is Tree Pruning?

Pruning is the process of shortening branches by converting some branch nodes into leaf nodes and removing leaf nodes under the original branch.

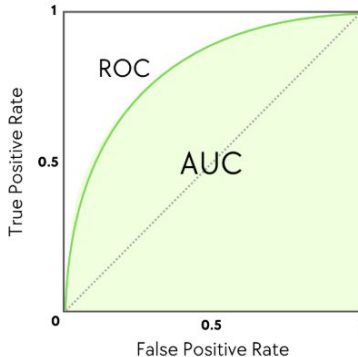
Key Goal: Reduce tree complexity while maintaining or improving predictive performance on new, unseen data.

- Simplifies the tree by removing unnecessary branches
- Reduces overfitting to training data
- Improves generalization on new data



Set 03: ROC curve, AUC

- **Main idea:** Receiver Operating Curve (ROC) shows the performance of the classification model at different threshold settings.
- ROC is a probability curve and AUC represents the degree or measure of separability.
- It tells how much the model is capable of distinguishing between classes.
- Higher the AUC, the better the model is at predicting 0 classes as 0 and 1 classes as 1.
- It is consider that Higher the AUC, the better the model is at distinguishing between patients with the disease and no disease.
- The ROC curve is plotted with True Positive Rate (TPR) against the False Positive Rate (FPR) where TPR is on the y-axis and FPR is on the x-axis.



Set 03: ROC curve, AUC

Example Problem-

- **Suppose**, a classification model yields the classification probability as shown in Table 1.

Step 1 - Let us set the different Threshold as $\alpha = [0, 0.2, 0.4, 0.8, 1]$

Step 2 - Taking first threshold as $\alpha = [0]$, we will get-

Step 3 - $TPR = \frac{4}{4+0} = 1$

& $FPR = \frac{2}{2+0} = 1$

Therefore TPR, FPR at $\alpha = [0]$ is (1,1)

Output	Class. Prob. (Y)	At $\alpha = [0]$ Y_1
1	0.8	1
0	0.96	1
1	0.4	1
1	0.3	1
0	0.2	1
1	0.7	1

Table 2. Predicted Output at $\alpha = [0]$

Output	Class. Prob. (Y)
1	0.8
0	0.96
1	0.4
1	0.3
0	0.2
1	0.7

Table 1. Actual Output vs Classification yielded Prob.

Set 03: ROC curve, AUC

Cntd...

Step 4 - Repeating all the steps from **Step 1** to **Step 3** at $\alpha = [0.2]$, we will get-

$$\text{TPR} = \frac{4}{4+0} = 1$$

$$\& \text{FPR} = \frac{1}{1+1} = 0.5$$

Therefore TPR, FPR at $\alpha = [0.2]$ is (1 , 0.5)

Step 5 - Again repeating all the steps from **Step 1** to **Step 3** at $\alpha = [0.4]$, we will get-

$$\text{TPR} = \frac{2}{2+2} = 0.5$$

$$\& \text{FPR} = \frac{1}{1+1} = 0.5$$

Therefore TPR, FPR at $\alpha = [0.4]$ is (0.5 , 0.5)

Output	Class. Prob. (Y)	At $\alpha = [0.2]$ Y_1
1	0.8	1
0	0.96	1
1	0.4	1
1	0.3	1
0	0.2	0
1	0.7	1

Table 3. Predicted Output at $\alpha = [0.2]$

Output	Class. Prob. (Y)	At $\alpha = [0.4]$ Y_1
1	0.8	1
0	0.96	1
1	0.4	0
1	0.3	0
0	0.2	0
1	0.7	1

Table 4. Predicted Output at $\alpha = [0.4]$

Set 03: ROC curve, AUC

Cntd...

Step 6 - Repeating all the steps from **Step 1** to **Step 3** at $\alpha = [0.8]$, we will get-

$$TPR = \frac{0}{0+4} = 0$$

$$\& FPR = \frac{1}{1+1} = 0.5$$

Therefore TPR, FPR at $\alpha = [0.8]$ is (0 , 0.5)

Step 7 - Again repeating all the steps from **Step 1** to **Step 3** at $\alpha = [1]$, we will get-

$$TPR = \frac{0}{0+4} = 0$$

$$\& FPR = \frac{0}{0+2} = 0$$

Therefore TPR, FPR at $\alpha = [1]$ is (0 , 0)

Output	Class. Prob. (Y)	At $\alpha = [0.8]$ Y_1
1	0.8	0
0	0.96	1
1	0.4	0
1	0.3	0
0	0.2	0
1	0.7	0

Table 5. Predicted Output at $\alpha = [0.8]$

Output	Class. Prob. (Y)	At $\alpha = [1]$ Y_1
1	0.8	0
0	0.96	0
1	0.4	0
1	0.3	0
0	0.2	0
1	0.7	0

Table 6. Predicted Output at $\alpha = [1]$

Set 03: ROC curve, AUC

Cntd...

Step 8 – Finally, plotting the **ROC** curve at different threshold setting we will get-

Threshold	TPR	FPR
0	1	1
0.2	1	0.5
0.4	0.5	0.5
0.8	0	0.5
1	0	0

Table 7. TPR, FPR values at different threshold settings.

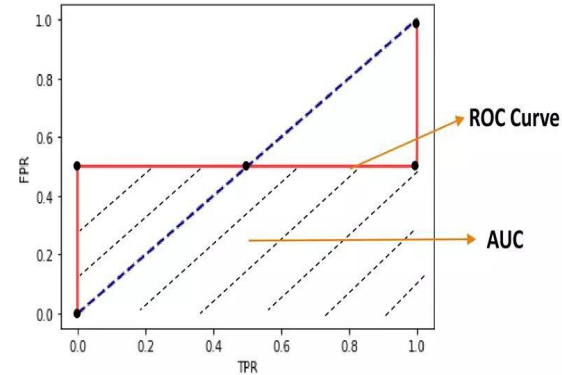
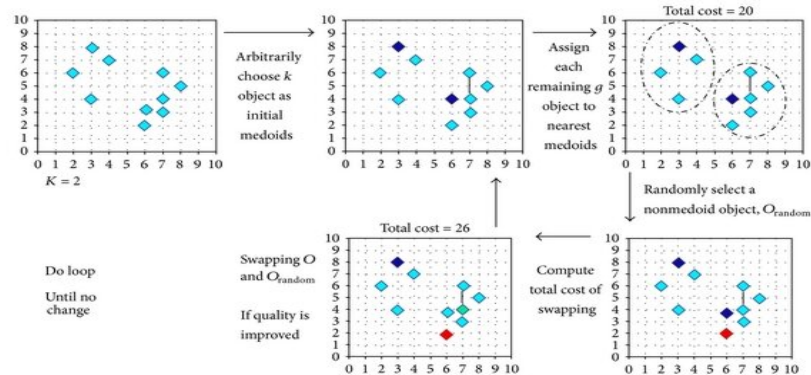


Fig. 2. ROC-AUC Curve.

Set 04: Cluster Analysis (K-Medoids)



1. Initially select **k random** points as the medoids from the given **n data points** of the data set.
2. **Associate each data point to the closest medoid** by using any of the most common **distance metrics**.
3. **Calculate the cost** as the total sum of the distances (also called dissimilarities) of the data points from the assigned medoid.
$$c = \sum_{C_i} \sum_{P_i \in C_i} |P_i - C_i|$$
4. **Swap one medoid point** with a **non-medoid point** and recalculate the cost.
5. If the **calculated cost** with the **new medoid point** is **more** than the previous cost, we **undo the swap**, and the algorithm converges else; **we repeat step 4**.

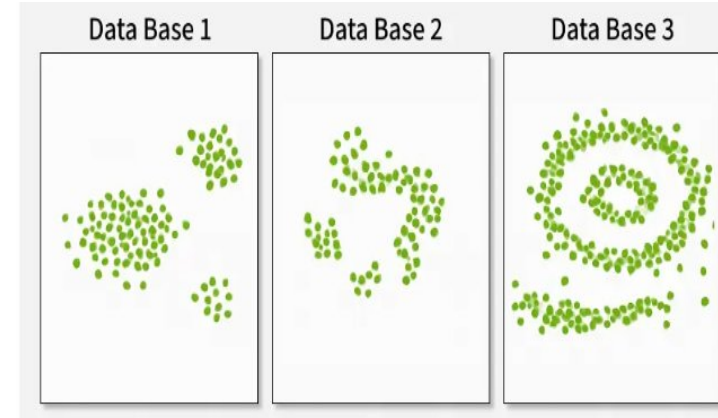
Set 05: Density Based Clustering (DBSCAN)

What is DBSCAN?

DBSCAN is a density-based clustering algorithm that groups data points that are closely packed together and marks outliers as noise based on their density in the feature space.

Key Advantage: Unlike K-Means or hierarchical clustering, DBSCAN does NOT assume clusters are compact and spherical—it can find arbitrary-shaped clusters and handle noise effectively.

- **Arbitrary-Shaped Clusters:** Clusters can take any shape, not just circular
- **Noise Handling:** Effectively identifies and isolates outliers
- **No Pre-defined K:** Automatically determines number of clusters



Key Parameters: *eps* and *MinPts*

eps (Epsilon)

Radius of neighborhood around each point

If *eps* too small → most points = noise

If *eps* too large → clusters merge together

MinPts (Minimum Points)

Min points within *eps* radius to form dense region

Rule of thumb: $\text{MinPts} \geq D+1$ (D =dimensions)

Minimum: $\text{MinPts} = 3$ recommended

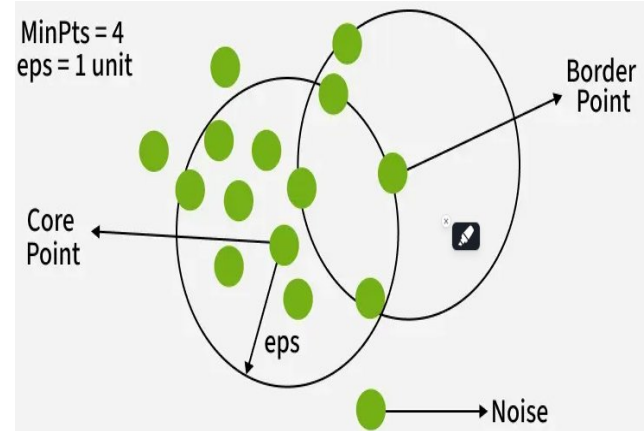
Set 05: Density Based Clustering (DBSCAN)

How Does DBSCAN Work?

DBSCAN works by categorizing data points into three types:

1. Core points which have a sufficient number of neighbors within a specified radius (epsilon)
2. Border points which are near core points but lack enough neighbors to be core points themselves
3. Noise points which do not belong to any cluster.

By iteratively expanding clusters from core points and connecting density-reachable points, DBSCAN forms clusters without relying on rigid assumptions about their shape or size.



THANKS
FOR
WATCHING
